

Two hidden layers compute the maximum of six numbers

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Abstract

We prove that $\max(x_1, \dots, x_6)$ is exactly computable by a ReLU neural network with two hidden layers of arbitrary width. This resolves an open problem of Bakaev, Brunck, Hertrich, Stade, and Yehudayoff, who established the five-input case and explicitly left the six-input case open, and which is also listed as open in subsequent work. We give an explicit construction as a signed combination of six S_6 -orbits of P_2 support functions with denominator 360, together with an exact, floating-point free computer-assisted proof of correctness. We show that such a representation must use cancellation — a signed Minkowski identity in which the simplex appears only after cancellation — for all $n \geq 5$. We then give computational evidence that six is the exact threshold: $\max_7$ and $\max_8$ are not representable in the same family even after enlarging it. Finally we isolate the obstruction to a separation as a global, cross-chamber phenomenon by a single-chamber invisibility lemma, and reduce a genuine two-versus-three hidden-layer separation to a single lattice normal-form conjecture.

1 Introduction

A feedforward ReLU network with k hidden layers computes a continuous piecewise-linear (CPWL) function, and conversely every CPWL function arises this way [1]. A central question is how the number of hidden layers limits expressivity, and the canonical test function is the maximum $\max_n(x) = \max(x_1, \dots, x_n)$. A balanced binary tree of pairwise maxima computes $\max_n$ with $\lceil \log_2 n \rceil$ hidden layers, and Hertrich, Basu, Di Summa, and Skutella [2] conjectured this is optimal: that exactly representing $\max_n$ requires $\lceil \log_2(n+1) \rceil$ hidden layers, which would make the ReLU depth hierarchy strict.

The conjecture holds for integer weights: Haase, Hertrich, and Loho [3] proved an integer-weight lower bound of $\lceil \log_2 n \rceil$ hidden layers via lattice-polytope volumes. Over the rationals the picture differs. Bakaev, Brunck, Hertrich, Stade, and Yehudayoff [4] showed that $\max_5$ needs only two hidden layers, disproving the $n = 5$ case and improving the general upper bound, and they left open whether two hidden layers suffice for six inputs.

Theorem 1. $\max(x_1, \dots, x_6)$ is computable by a ReLU network with two hidden layers.

The construction uses rational weights, as it must: the integer-weight lower bound [3] would otherwise force three layers. Our proof is computer-assisted but exact; every arithmetic step is over the integers or rationals, and no floating-point computation enters the certificate.

2 Preliminaries

For a polytope $P \subset \mathbb{R}^n$ its support function is $h_P(x) = \max_{g \in P} \langle g, x \rangle$. Then $\max_n = h_\Delta$ for the standard simplex $\Delta = \text{conv}\{e_1, \dots, e_n\}$. Support functions are additive under Minkowski sum,

$h_{P+Q} = h_P + h_Q$, and satisfy $h_{\text{conv}(A \cup B)} = \max(h_A, h_B)$; two polytopes with equal support functions are equal.

Following [4], define $P_0 = \{\text{points}\}$ and $P_{k+1} = \{\sum_i P_i * Q_i : P_i, Q_i \in P_k\}$, where $*$ is join (convex hull of union) and the sum is Minkowski. Then P_1 is the class of zonotopes, and P_2 is Minkowski sums of *joins of two zonotopes*. A *block* is $R = \text{conv}(Z_1 \cup Z_2)$ with Z_1, Z_2 zonotopes, and $h_R(x) = \max(h_{Z_1}(x), h_{Z_2}(x))$.

Lemma 1 (Two-layer dictionary). *A CPWL function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is computable by a ReLU network with two hidden layers if and only if $f = \sum_t c_t h_{R_t} + \ell$ for finitely many blocks R_t , real coefficients c_t , and an affine-linear ℓ . In particular $\max_6 \in$ two hidden layers iff $h_\Delta = \sum_t c_t h_{R_t} + \ell$ for blocks R_t .*

Proof. By [1, 3] a function is k -hidden-layer computable iff it is a difference $h_Q - h_P$ of support functions of $P, Q \in P_k$ (virtual polytopes), up to an affine term. For $k = 2$, P and Q are Minkowski sums of blocks, so by additivity $h_P = \sum_i h_{A_i}$ and $h_Q = \sum_j h_{B_j}$ over blocks A_i, B_j ; then $f = h_Q - h_P + \ell = \sum_j h_{B_j} - \sum_i h_{A_i} + \ell$ is a signed sum of block support functions. Conversely any signed sum $\sum_t c_t h_{R_t}$ is $h_Q - h_P$ with $Q = \sum_{c_t > 0} c_t R_t$ and $P = \sum_{c_t < 0} (-c_t) R_t$ in P_2 , hence two-hidden-layer. The affine ℓ is realized in the output layer. $\square$

The weight-2 lattice. We take building-block vertices on the weight-2 lattice points of the dilated simplex $2\Delta^5$, namely $\{2e_i\}$ and $\{e_i + e_j\}$ (21 points in $\mathbb{R}^6$); zonotopes are generated by braid directions $e_a - e_b$ restricted so that all vertices stay on the lattice. This is the lattice underlying the construction of [4] for $\max_5$. Write $V_2^{(2)}$ for the (finite) family of signed sums of block support functions with all block vertices in this lattice.

Lemma 2 (Symmetrization). *If $h_\Delta = \sum_t c_t h_{R_t} + \ell$ with blocks R_t , then also*

$$h_\Delta = \sum_t c_t \left(\frac{1}{|S_n|} \sum_{\sigma \in S_n} h_{\sigma R_t} \right) + \bar{\ell},$$

an S_n -symmetric representation in which each term is an orbit sum of blocks and $\bar{\ell}$ is a scalar multiple of $\sum_i x_i$. Consequently $\max_n \in$ two layers iff it has an S_n -symmetric such representation, and the search may be restricted to S_n -orbits of blocks (for $n = 6$ this collapses 16390 two-way joins to 78 orbit classes, making an exact linear solve tractable).

Proof. For $\sigma \in S_n$, $h_{\sigma R}(x) = h_R(\sigma^{-1}x)$, and h_Δ is S_n -invariant ($h_\Delta(\sigma^{-1}x) = h_\Delta(x)$, since σ permutes the vertices of Δ). Averaging the identity $h_\Delta = \sum_t c_t h_{R_t} + \ell$ over $\sigma \in S_n$ gives $h_\Delta = \sum_t c_t \overline{h_{R_t}} + \bar{\ell}$, where $\overline{h_{R_t}} = \frac{1}{|S_n|} \sum_{\sigma} h_{\sigma R_t}$. Each σR_t is again a block (σ maps zonotopes to zonotopes), so $\overline{h_{R_t}}$ is a nonnegative combination of block support functions and the whole representation remains a signed sum of block support functions (Lemma 1); equivalently $\overline{h_{R_t}} = \frac{1}{|S_n|} h_{\sum_{\sigma} \sigma R_t}$ is a positive multiple of the support function of the S_n -symmetric Minkowski orbit sum $\sum_{\sigma} \sigma R_t$. For the linear part, write $\ell(x) = \langle a, x \rangle + b$; then $\bar{\ell}(x) = \langle \bar{a}, x \rangle + b$ with $\bar{a} = \frac{1}{|S_n|} \sum_{\sigma} \sigma a = \frac{1}{n} (\sum_i a_i) \mathbf{1}$, so $\bar{\ell}$ is a multiple of $\sum_i x_i$ plus the constant b . Evaluating the original identity at $x = 0$ gives $b = 0$ (support functions vanish at the origin), so $\bar{\ell}$ is a scalar multiple of $\sum_i x_i$. $\square$

3 The construction

Solving exactly for orbit coefficients reproducing $\max_6$ and minimizing the number of terms yields a construction with six orbit terms and denominator $D = 360$:

$$\max_6 = \sum_{o=1}^6 c_o \left(\sum_{R \in o} h_R \right), \quad c_o \in \left\{ \frac{1}{360}, -\frac{1}{360}, \frac{1}{360}, \frac{1}{90}, -\frac{1}{90}, -\frac{1}{180} \right\},$$

with zero linear part. The six orbit representatives, as vertex (gradient) sets, are listed in the accompanying file `construction_max6.txt`. Each R is a join of two weight-2 zonotopes, so each h_R is a two-hidden-layer function and the signed sum is two hidden layers. Expanding the six orbits over S_6 gives 1530 individual P_2 terms.

The construction uses negative coefficients, and this is unavoidable: the representation is a genuine *virtual* (signed) Minkowski identity in which the simplex appears only after cancellation.

Proposition 1 (Necessity of cancellation). *For $n \geq 5$, every representation of $\max_n$ as a signed sum $\sum_t c_t h_{R_t} + \ell$ of support functions of two-way joins of zonotopes (with ℓ linear) has at least one negative coefficient.*

Proof. Suppose all $c_t \geq 0$. Write $\ell(x) = \langle a, x \rangle + b$; evaluating the identity $h_\Delta = \sum_t c_t h_{R_t} + \ell$ at $x = 0$ forces $b = 0$ (support functions vanish at the origin), so $\ell = h_{\{a\}}$ and additivity gives $h_\Delta = h_{a + \sum_t c_t R_t}$, whence $\Delta = a + \sum_t c_t R_t$ as a Minkowski sum. For each t , grouping the identity as $\Delta = (c_t R_t) + (a + \sum_{s \neq t} c_s R_s)$ exhibits $c_t R_t$ as a Minkowski summand of Δ ; since a simplex of dimension ≥ 2 is Minkowski indecomposable up to homothety and translation (Shephard; McMullen), each nonzero $c_t R_t$ is homothetic to Δ , i.e. each nonzero R_t is a homothet of Δ .

It remains to show that a homothet of the $(n-1)$ -simplex is a two-way join of zonotopes only for $n \leq 4$. Let $S = \text{conv}(Z_1 \cup Z_2)$ be an $(n-1)$ -simplex with Z_1, Z_2 zonotopes. Each Z_i is centrally symmetric about some center c_i . If a vertex v of S is a vertex of Z_i , then by central symmetry $2c_i - v \in Z_i \subseteq S$; in barycentric coordinates on S with $v = e_j$, membership $2c_i - v \in S$ forces the j -th barycentric coordinate to be nonnegative, i.e. $2(c_i)_j - 1 \geq 0$, so $(c_i)_j \geq \frac{1}{2}$. Hence at most two vertices of S can be vertices of Z_i : three would give three barycentric coordinates of c_i each $\geq \frac{1}{2}$, summing to more than 1, impossible. Now every vertex of S is an extreme point of $Z_1 \cup Z_2$, hence a vertex of Z_1 or of Z_2 ; so S has at most $2 + 2 = 4$ vertices, i.e. $n \leq 4$, contradicting $n \geq 5$. (The bound is sharp: for $n = 4$ a tetrahedron is the join of two skew segments.) $\square$

For $n = 4$ a single term (no negativity) suffices; $\max_5$ and $\max_6$ require cancellation. This is why single-polytope invariants (central symmetry, volume) do not obstruct: the obstruction lives in the cancellation, not in any individual body.

4 The proof of Theorem 1

Let $C = \sum_o c_o (\sum_{R \in o} h_R)$ denote the construction of the previous section, a CPWL function. We must show $C = h_\Delta = \max_6$ on all of $\mathbb{R}^6$. We proceed by three lemmas and two machine-checked computational facts; every step below is exact (integer or rational), with no floating-point arithmetic.

Lemma 3 (Gradient reduction). *C and $\max_6$ are CPWL. If $\nabla C = e_1$ at an interior point of every full-dimensional cell of the weight-2 difference arrangement inside the open fundamental cone $\{x_1 > \dots > x_6\}$, then $C = \max_6$ on all of $\mathbb{R}^6$.*

Proof. On any region where no block changes its maximizing vertex, $\nabla C = \sum_t c_t g_t^*$ with $g_t^* = \arg \max_{g \in R_t} \langle g, x \rangle$; the selection g_t^* is locally constant and changes only across the hyperplanes $\langle g - g', x \rangle = 0$ for g, g' vertices of a common block (the weight-2 difference hyperplanes), so these hyperplanes contain all non-smooth loci of C . Inside a cell every relevant comparison is strict, so each g_t^* is unique and ∇C is well defined and constant. On the open fundamental cone $\max_6 = x_1$ so $\nabla \max_6 = e_1$. Suppose $\nabla C = e_1$ at an interior point of every cell. Since C is affine on each cell, $\nabla C = e_1$ throughout each open cell, so $C - x_1$ is constant on each cell; by continuity of the CPWL function $C - x_1$ across a shared wall, the constants on two adjacent cells agree, and since the cell-adjacency graph of the arrangement inside the connected open cone is connected (Lemma 5), $C - x_1$ equals a single constant κ on the open cone, hence on its closure. As C and x_1 are positively homogeneous of degree 1, so is $C - x_1$; thus $\kappa = (C - x_1)(\lambda x) = \lambda \kappa$ for all $\lambda > 0$, forcing $\kappa = 0$. Therefore $C = x_1 = \max_6$ on the closed fundamental chamber. Both C (an orbit-summed signed sum) and $\max_6$ are S_6 -invariant, and $S_6 \cdot \{x_1 \geq \dots \geq x_6\} = \mathbb{R}^6$, so equality on one chamber gives equality on all of $\mathbb{R}^6$. $\square$

Lemma 4 (Forced directions). *In gap coordinates $y_k = x_k - x_{k+1} \geq 0$ ($k = 1, \dots, 5$), a difference hyperplane with normal h becomes $\sum_k (\sum_{i \leq k} h_i) y_k = 0$. If all coefficients $\sum_{i \leq k} h_i$ have one sign, the hyperplane meets the open cone $y > 0$ trivially and cannot subdivide it. Of the 120 distinct weight-2 difference hyperplanes, exactly 85 are so forced; the cell structure inside $y > 0$ is therefore the arrangement of the remaining 35 hyperplanes in $\mathbb{R}^5$.*

Proof. The change of coordinates $x \mapsto y$ is linear with $x_i - x_j = \sum_{i \leq k < j} y_k$; substituting gives the stated form. If every prefix sum $\sum_{i \leq k} h_i$ is ≥ 0 (resp. ≤ 0), the linear form is ≥ 0 (resp. ≤ 0) on $y \geq 0$ and vanishes on $y > 0$ nowhere, so the hyperplane does not cut the open cone. Counting the forced normals among the 120 difference directions leaves 35 unforced; only these subdivide $\{y > 0\}$. (The counts 85 and 35 are verified by exact integer arithmetic in `check_max6.py`.) $\square$

Computational Fact 1 (enumeration). A deterministic depth-first search over sign vectors of the 35 unforced normals, pruning branches whose open sign system is infeasible, enumerates exactly 2608 full-dimensional cells of $\{y > 0\}$, each with an explicit exact integer interior witness.

Computational Fact 2 (gradient). At each of the 2608 integer witnesses, the exact-integer gradient $\sum_t c_t g_t^* = e_1$.

Both facts are established by exact integer computation in `check_max6.py`. As an independent check, evaluating C and $\max_6$ in exact rational arithmetic at random points — including boundary and tie configurations — confirms $C = \max_6$ with zero error. It remains only to certify that the enumeration is complete.

Lemma 5 (Completeness by adjacency closure). *Let $\mathcal{C}$ be the set of 2608 enumerated cells. If for every cell in $\mathcal{C}$ and every single-coordinate sign flip the resulting open sign system is either another cell of $\mathcal{C}$ or certifiably empty, then $\mathcal{C}$ is the set of all full-dimensional cells of the arrangement inside $\{y > 0\}$.*

Proof. The full-dimensional cells of a hyperplane arrangement inside the connected open cone $\{y > 0\}$ form the vertices of a graph in which two cells are adjacent iff their (closed) sign vectors differ in exactly one coordinate; this graph is connected (any two points of the cone are joined by a path crossing one wall at a time). $\mathcal{C}$ is nonempty and, by hypothesis, closed under adjacency (each neighbor is in $\mathcal{C}$ or empty, i.e. not a cell). A nonempty subset of a connected graph closed under adjacency is the whole vertex set, so $\mathcal{C}$ contains every cell. Emptiness of a flipped sign system

$\{y : \langle g_j, y \rangle > 0 \ \forall j\}$ is certified exactly by Gordan's theorem: the open system is infeasible iff there exist nonnegative rationals x_j , not all zero, with $\sum_j x_j g_j = 0$ (over the orthant generators and the signed normals); such an x is exhibited and the identity checked over $\mathbb{Q}$. Across all 2608×35 flips every neighbor is in-set (13756) or Gordan-certified empty (77524), none uncertified. $\square$

Proof of Theorem 1. By Lemma 4 the cells of the fundamental cone are those of the 35-hyperplane arrangement. By Computational Fact 1 and Lemma 5 the 2608 enumerated cells are *all* of them. By Computational Fact 2, $\nabla C = e_1$ at an interior witness of each. By Lemma 3, $C = \max_6$ on $\mathbb{R}^6$. Since C is a signed sum of block support functions, Lemma 1 makes it a two-hidden-layer ReLU function. $\square$ $\square$

5 The threshold and a route to a separation

The two-hidden-layer functions whose building blocks have vertices in the weight-2 lattice form a finite family: a weight-2 zonotope has vertices among the finitely many lattice points, and the building blocks are the two-way joins of such zonotopes (including degenerate point zonotopes, giving pyramids). We enumerate this family completely (all zonotopes generated by arbitrary, not merely braid, weight-2 point differences) and solve over $\mathbb{Q}$.

Proposition 2. *$\max_7$ is not a signed sum of support functions of two-way joins of weight-2 zonotopes. Equivalently, $\max_7$ has no two-hidden-layer representation with all building-block vertices in the weight-2 lattice.*

This is exact: the complete family yields a linear system over $\mathbb{Q}$ that is inconsistent for $\max_7$ (and consistent for $\max_6$, recovering Theorem 1). The same holds for $\max_8$. This locates a qualitative change exactly at $n = 6$, and we conjecture:

Conjecture 1. *$\max_n$ is computable in two hidden layers if and only if $n \leq 6$.*

Proposition 2 is a genuine lower bound over the weight-2 lattice; what is missing for an unconditional separation is only that the lattice may be assumed weight-2. The reduction $\max_n$ two-layer $\iff \Delta + P = Q$ with $P, Q \in P_2$ is standard; one checks that any representation may be symmetrized, and that $\Delta + P = Q$ forces the normal fan of Q to refine the braid fan. The remaining content is a normal-form lemma:

Conjecture 2 (Lattice normal form). *If $\max_n$ is computable in two hidden layers, it has a representation with all building-block vertices in the weight-2 lattice.*

Granting this, two-layer representability becomes a finite computation, and the $n = 7$ negative becomes the statement that $\max_7$ requires three hidden layers, a separation of two from three hidden layers over the reals.

The difficulty of the normal-form lemma has a clean structural source: the obstruction is invisible inside any single chamber of the braid arrangement. Let $D = \{x_1 \geq \dots \geq x_n\}$ be the closed fundamental chamber; on D one has $\max_n = x_1$, a linear function.

Lemma 6 (Single-chamber invisibility). *Let λ be a finite signed combination of point evaluations, all at points of a single open braid chamber, and suppose λ annihilates every linear function. Then $\lambda(\max_n) = 0$. Consequently, any separating functional — one with $\lambda(h_R) = 0$ for all blocks R , $\lambda(\ell) = 0$ for all linear ℓ , and $\lambda(\max_n) \neq 0$ — cannot have all its evaluation points in a single open braid chamber; it must draw on at least two chambers.*

Proof. On one open braid chamber $\{x_{\sigma(1)} > \dots > x_{\sigma(n)}\}$, the largest coordinate is $x_{\sigma(1)}$ at every point (the ordering index σ is constant on the chamber), so $\max_n$ agrees there with the single linear function $\ell^*(x) = x_{\sigma(1)}$. Since all evaluation points lie in that chamber and λ annihilates linear functions, $\lambda(\max_n) = \sum_r a_r \max_n(z_r) = \sum_r a_r \ell^*(z_r) = \lambda(\ell^*) = 0$. The second statement is the contrapositive: if $\lambda(\max_n) \neq 0$, its evaluation points cannot all lie in one open chamber. $\square$

Thus the membership obstruction is a genuinely global, cross-chamber compatibility failure: the local linear pieces of $\max_n$ exist chamber by chamber but need not glue to a symmetric P_2 representation. This is borne out at the level of certificates: the exact rational functional witnessing Proposition 2 at weight 2 has minimal support rank+1 spread across almost as many distinct braid chambers, with no sparse or single-chamber representative, exactly as Lemma 6 predicts. It also explains why single-chamber or local-linear tests (and, in our experiments, low-order finite differences, wall relaxations, and mixed-volume comparisons) do not obstruct: they cannot see a cross-chamber phenomenon. (We caution that the Lemma rules out single-chamber certificates only; adjacent-chamber and higher-order tests can still detect finite-family obstructions.)

The route this suggests is a global one. We note first that the conformity route is closed: the optimal shallow construction is provably not braid-conforming (indeed our $\max_6$ construction requires non-conforming blocks), consistent with the conditional bounds of Grillo, Hertrich, and Loho [6]. The relevant tools for the lattice normal form are the decomposition polyhedra of Brandenburg, Grillo, and Hertrich [5] — the natural global object, in which two-layer representations are extreme decompositions and non-representability is emptiness of the P_2 -restricted polyhedron — and deformation-cone theory for the braid fan. Contemporary lower bounds are consistent with Theorem 1: [3] is integer-weight only, [7] is for finite-precision weights, [6] is for braid-conforming networks, and [8] is a width rather than depth bound at depth at least three.

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